SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2024 AI&DS	**Date:**	26-10-2026

Code of Conduct

*I _____**P.Sunayana(192424303)**_____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: _**P.Sunayana**_*

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Dr Veeramani		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Fundamentals of Mobile Computing
2. Evolution of Cellular Networks (1G to 5G)
3. Cellular Network Architecture
4. Cell Structure and Cellular Concept
5. Frequency Reuse in Cellular Networks
6. Co-channel and Adjacent Channel Interference
7. Cell Splitting and Cell Sectoring Techniques
8. Cellular Capacity Enhancement Techniques
9. Access Channels in Mobile Communication
10. Mobile Station (MS) Architecture and Functions
11. Base Station Subsystem (BSS) Architecture
12. Network Switching Subsystem (NSS) Operations
13. Mobile Network Protocol Stack
14. Core Network Architecture in Cellular Systems
15. Public Land Mobile Network (PLMN) Architecture
16. Mobility Management in Cellular Networks
17. Handoff Mechanisms and Algorithms
18. Roaming in Mobile Communication Networks
19. Location Management Techniques
20. Call Setup and Call Termination Procedures
21. Paging and Registration in Cellular Networks
22. Cellular System Operations and Signaling
23. Performance Metrics of Cellular Networks
24. Traffic Management in Cellular Systems
25. Radio Resource Management Techniques
26. Network Planning and Cell Deployment Strategies
27. Quality of Service (QoS) in Mobile Networks
28. Mobility Models in Mobile Computing
29. Challenges in Mobile Computing and Cellular Networks
30. Future Trends in Mobile Computing and Intelligent Cellular Networks
31. Wireless Channel Characteristics and Propagation Models
32. Multipath Propagation, Fading and Shadowing in Mobile Networks

33. Doppler Effect and Coherence Time in Wireless Communication
34. Modulation and Coding Techniques for Mobile Communication
35. Multiple Access Techniques: FDMA, TDMA, CDMA and OFDMA
36. Mobile Communication Antennas and Diversity Techniques
37. Wireless Link Adaptation and Adaptive Modulation
38. Energy-Efficient Communication in Mobile Devices and Networks
39. Security and Authentication Mechanisms in Cellular Networks
40. Heterogeneous Wireless Networks and Interworking Technologies

Cell Structure and Cellular Concept

Presented by :
P.Sunayana (192424303)

The Cellular Concept

Replace one high-power transmitter with many low-power ones, each covering a small area called a cell.

1

Frequency reuse

Same channels reused in distant cells

2

Higher capacity

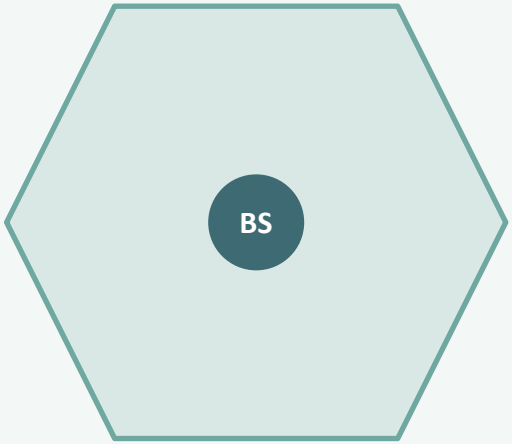
More users with limited spectrum

3

Low power

Smaller, lighter handsets and batteries

Cell Structure



Hexagon: ideal model with no gaps or overlap

Base Station (BS)

Antenna and transceiver at the cell centre

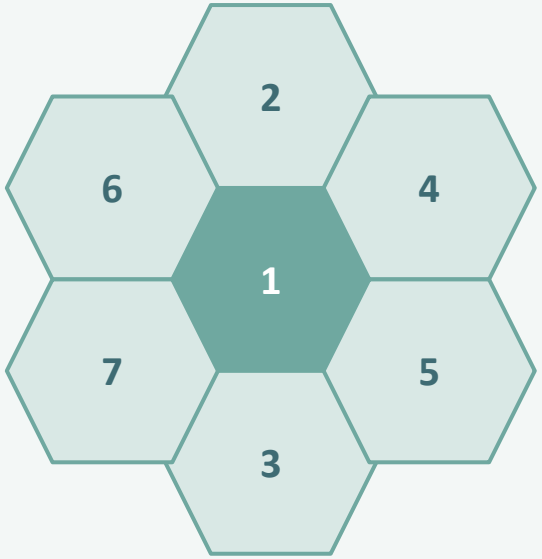
Mobile Station (MS)

The user's handset inside the cell

Mobile Switching Centre

Links cells to each other and the PSTN

Frequency Reuse and Clusters



Cluster of N = 7 cells

Key relations

- Cluster size: $N = i^2 + ij + j^2$
- Reuse distance: $D = R\sqrt{(3N)}$
- Reuse ratio: $Q = D / R = \sqrt{(3N)}$
- Smaller N gives more capacity but more interference

Improving Capacity

Cell Splitting

Divide a congested cell into smaller cells with lower-power base stations.

- Radius reduced, often by half
- More channels per unit area
- Suits growing urban demand

Sectoring

Replace one omnidirectional antenna with directional antennas (e.g. 120°).

- Cuts co-channel interference
- Improves signal quality
- Needs more handoffs

Handoff and Interference

Handoff

- Call moves to a new base station as the user crosses cells
- Triggered when signal falls below a threshold
- Must be seamless to avoid dropped calls

Interference

- Co-channel: from cells reusing the same frequency
- Adjacent channel: from neighbouring frequencies
- Reduced by larger D/R and good channel planning

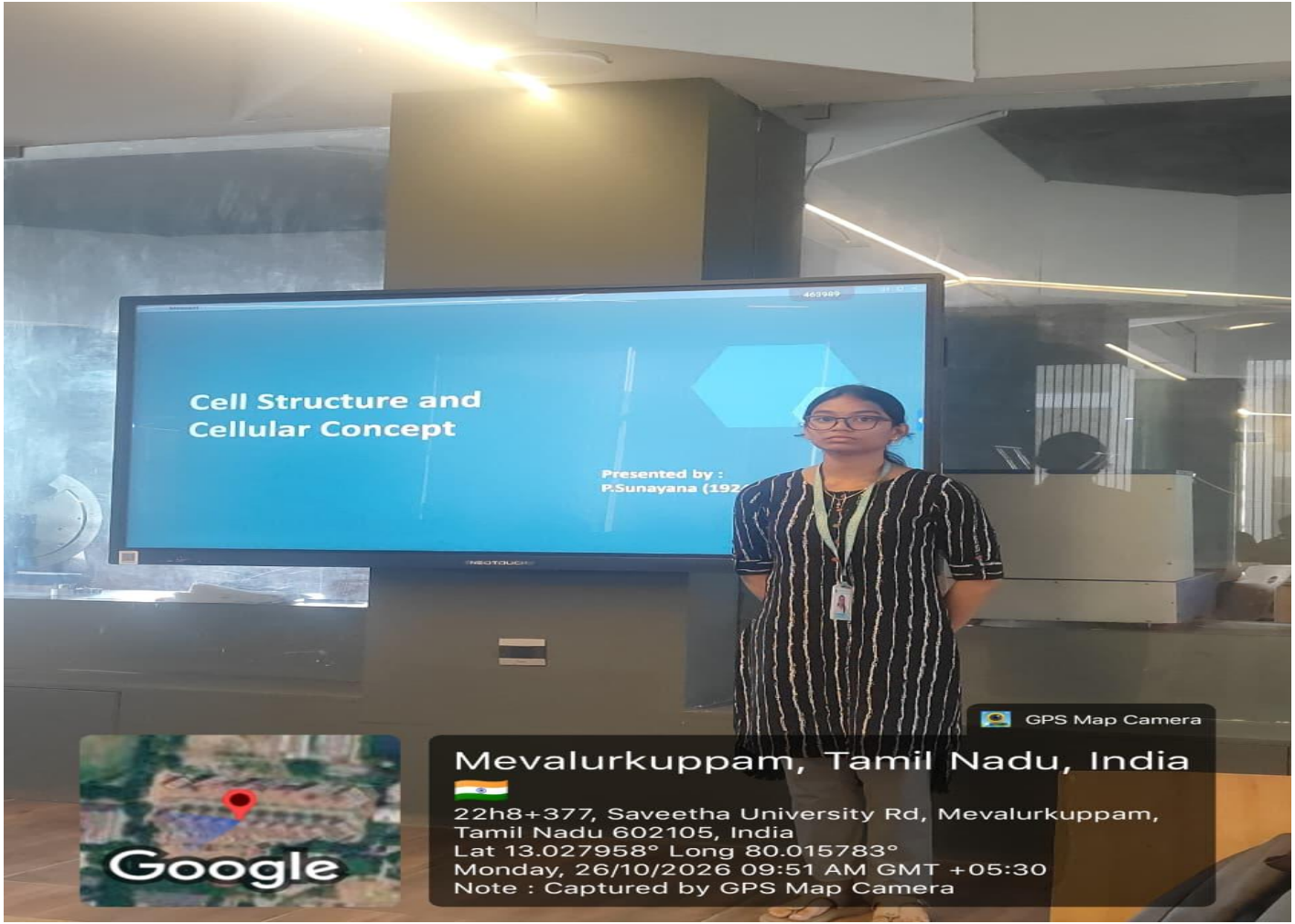
Key Takeaways

1. Cells are small coverage areas, modelled as hexagons
2. Frequency reuse across clusters multiplies capacity
3. Cluster size N sets the trade-off between capacity and interference
4. Cell splitting and sectoring add capacity further
5. Handoff keeps calls continuous as users move

Conclusion

In conclusion, the cellular concept divides a large geographical area into smaller cells to provide efficient mobile communication. Frequency reuse increases network capacity, while cell splitting and sectoring improve capacity and reduce interference. Handoff ensures continuous communication as users move between cells. Thus, these techniques help provide reliable, efficient, and high-quality mobile communication services.

THANK YOU


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Cellular Concept
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